

X-ray Diffraction from Crystals

Path difference, structure factor, and systematic absences

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Geometry and assumptions

Let the incident and scattered wavevectors be $\mathbf{k}_i$ and $\mathbf{k}_f$, with $|\mathbf{k}_i| = |\mathbf{k}_f| = k = 2\pi/\lambda$. Define the scattering vector

$$\mathbf{q} = \mathbf{k}_f - \mathbf{k}_i.$$

- Elastic scattering: $|\mathbf{k}_i| = |\mathbf{k}_f|$.
- A lattice point is $\mathbf{R}_{mn} = m\mathbf{a}_1 + n\mathbf{a}_2$.
- The atomic structure factor is taken angle-independent:
 $f_j(\theta) = f_j$.

For a symmetric reflection, the angle between the incident beam and the lattice planes is θ , hence the angle between the incident and diffracted beams is 2θ .

Path difference

For atoms at $\mathbf{r}$ and $\mathbf{r} + \mathbf{R}$, the extra distance travelled by the incident ray is $-\hat{\mathbf{k}}_i \cdot \mathbf{R}$, while that on exit is $+\hat{\mathbf{k}}_f \cdot \mathbf{R}$. Therefore

$$\Delta(\mathbf{R}) = (\hat{\mathbf{k}}_f - \hat{\mathbf{k}}_i) \cdot \mathbf{R}$$

and the phase difference is

$$\phi(\mathbf{R}) = \frac{2\pi}{\lambda} \Delta(\mathbf{R}) = \mathbf{q} \cdot \mathbf{R}.$$

For $\mathbf{R}_{mn} = m\mathbf{a}_1 + n\mathbf{a}_2$,

$$\phi_{mn} = m\mathbf{q} \cdot \mathbf{a}_1 + n\mathbf{q} \cdot \mathbf{a}_2$$

Constructive interference occurs when ϕ_{mn} is an integer multiple of 2π for every lattice translation.

Laue condition and Bragg form

The intensity is maximized when

$$\mathbf{q} \cdot \mathbf{a}_1 = 2\pi h, \quad \mathbf{q} \cdot \mathbf{a}_2 = 2\pi k,$$

where $h, k \in \mathbb{Z}$. In three dimensions this is

$$\mathbf{q} = \mathbf{G}_{hkl} = h\mathbf{b}_1 + k\mathbf{b}_2 + l\mathbf{b}_3,$$

with $\mathbf{a}_i \cdot \mathbf{b}_j = 2\pi\delta_{ij}$.

$$|\mathbf{G}_{hkl}| = \frac{2\pi}{d_{hkl}}, \quad |\mathbf{q}| = 2k \sin \theta = \frac{4\pi}{\lambda} \sin \theta.$$

Thus

$$\boxed{2d_{hkl} \sin \theta = n\lambda}$$

which is Bragg's law. The integer n can be absorbed into the Miller indices, so the first-order reciprocal-lattice condition is normally used.

Structure factor

Let the unit cell contain atoms $j = 1, \dots, N$, at fractional coordinates (x_j, y_j, z_j) , with $\mathbf{r}_j = x_j \mathbf{a}_1 + y_j \mathbf{a}_2 + z_j \mathbf{a}_3$. The scattered amplitude is

$$A(\mathbf{q}) = \sum_{\mathbf{R}} \sum_j f_j \exp[i\mathbf{q} \cdot (\mathbf{R} + \mathbf{r}_j)].$$

At a reciprocal-lattice point, $\exp(i\mathbf{q} \cdot \mathbf{R}) = 1$, and the unit-cell amplitude is the structure factor

$$F_{hkl} = \sum_{j=1}^N f_j \exp\{2\pi i(hx_j + ky_j + lz_j)\}.$$

The diffracted intensity is

$$I_{hkl} \propto |F_{hkl}|^2.$$

If all atoms are identical with constant f , then

$$F_{hkl} = f \sum_j e^{2\pi i(hx_j + ky_j + lz_j)}.$$

Simple cubic (SC)

The conventional SC cell has one lattice point:

$$(0, 0, 0).$$

Therefore

$$F_{hkl} = f, \quad I_{hkl} \propto f^2.$$

Extinction rule

There are no systematic absences: all integer (hkl) are allowed, apart from the trivial exclusion (000) as a Bragg reflection.

Body-centred cubic (BCC)

Basis positions:

$$(0, 0, 0), \quad \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right).$$

Hence

$$F_{hkl} = f \left[1 + e^{\pi i(h+k+l)} \right] = f \left[1 + (-1)^{h+k+l} \right].$$

Extinction rule

$h + k + l = 2n$: allowed; $h + k + l = 2n + 1$: absent.

Face-centred cubic (FCC)

Basis positions:

$$(0, 0, 0), (0, \frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, 0, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}, 0).$$

Thus

$$F_{hkl} = f [1 + (-1)^{k+l} + (-1)^{h+l} + (-1)^{h+k}].$$

Extinction rule

Reflections occur only when h, k, l are all even or all odd. Mixed parity gives $F_{hkl} = 0$.

For allowed reflections, $|F| = 4f$.

Hexagonal close-packed (HCP)

For the primitive hexagonal cell, use the two-point basis

$$(0, 0, 0), \quad \left(\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right).$$

The basis factor is

$$F_{hkl} = f \left[1 + e^{2\pi i(2h+k)/3} e^{\pi i l} \right].$$

Equivalently,

$$F_{hkl} = f \left[1 + (-1)^l e^{2\pi i(2h+k)/3} \right].$$

Extinction rule (ideal HCP basis)

The basis factor vanishes only when l is odd and $2h + k \equiv 0 \pmod{3}$. For even l , it is nonzero for every h, k .

Diamond cubic (DC)

Diamond cubic is an FCC lattice plus the basis translation

$$\mathbf{t} = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right).$$

Its structure factor factors as

$$F_{hkl} = f S_{\text{FCC}}(hkl) \left[1 + e^{\frac{i\pi}{2}(h+k+l)} \right],$$

where $S_{\text{FCC}} = 1 + (-1)^{k+l} + (-1)^{h+l} + (-1)^{h+k}$.

Extinction rule

First require FCC parity: h, k, l all even or all odd. If all even, reflections with $h + k + l \equiv 2 \pmod{4}$ are absent. If all odd, the diamond factor is nonzero; hence all-odd reflections are allowed.

Examples: Si allows 111, 220, and 400; 222 is absent.

One-page rule sheet

Lattice	Systematic condition for nonzero F_{hkl}
SC	All hkl allowed.
BCC	$h + k + l$ even.
FCC	h, k, l all even or all odd.
HCP basis	Nonzero except when l is odd and $2h + k \equiv 0 \pmod{3}$.
Diamond cubic	FCC condition plus, for all-even indices, $h + k + l \not\equiv 2 \pmod{4}$.

Important distinction

These are systematic absences caused by translational symmetry. Additional zeros can arise from the actual atomic form factors, multiple chemical species, disorder, thermal motion, or non-ideal atomic positions.

Conventions and checks

- A different origin changes F_{hkl} by an overall phase, not the intensity $|F_{hkl}|^2$.
- HCP extinction statements depend on the chosen hexagonal basis and on whether one is describing the ideal two-atom basis or a larger conventional cell.
- For a multi-element crystal, replace the common f by the appropriate f_j in the general structure-factor sum.
- The angle-independent-atom assumption isolates geometric interference; real X-ray form factors generally depend on angle through $\sin \theta / \lambda$.

BCC selection rules

Plane	N	Plane	N	Plane	N
100	1	221	9	411, 330	18
110	2	310	10	331	19
111	3	311	11	420	20
200	4	222	12	421	21
210	5	320	13	332	22
211	6	321	14	422	24
220	8	400	16	431	26
		410	17	432	29

Allowed reflections shown in orange

For BCC, $h + k + l$ must be even; equivalently, the allowed values of $N = h^2 + k^2 + l^2$ in this range are even.

FCC selection rules

Plane	N	Plane	N	Plane	N
100	1	221	9	411, 330	18
110	2	310	10	331	19
111	3	311	11	420	20
200	4	222	12	421	21
210	5	320	13	332	22
211	6	321	14	422	24
220	8	400	16	431	26
		410	17	432	29

Allowed reflections shown in orange

For FCC, h, k, l must be all even or all odd; mixed parity is systematically absent.

Diamond cubic selection rules

Plane	N	Plane	N	Plane	N
100	1	221	9	411, 330	18
110	2	310	10	331	19
111	3	311	11	420	20
200	4	222	12	421	21
210	5	320	13	332	22
211	6	321	14	422	24
220	8	400	16	431	26
		410	17	432	29

Allowed reflections shown in orange

Diamond cubic first obeys FCC parity; all-even reflections with $h + k + l \equiv 2 \pmod{4}$ are additionally absent.

$\sin^2 \theta$ ratios for allowed reflections

SC 1 : 2 : 3 : 4 : 5 : 6 : 8 : ...

BCC 1 : 2 : 3 : 4 : 5 : 6 : 7 : ...

FCC 3 : 4 : 8 : 11 : 12 : ...

DC 3 : 8 : 11 : 16 : ...

How to use the sequence

For a cubic crystal, $\sin^2 \theta = (\lambda^2/4a^2)N$, where $N = h^2 + k^2 + l^2$. Each row is shown in its conventional integer scale; the BCC row is the allowed N sequence divided by 2.

Questions from lecture slide 21

1. The $\sin^2 \theta$ values calculated from the peak positions of an x-ray powder diffraction of an element were as follows: 0.120, 0.238, 0.357, 0.475, 0.593, 0.711, 0.83. Copper $K\alpha$ radiation was used. Determine the crystal structure and the lattice parameter.
2. Only two reflections are observed on the powder pattern of a cubic crystal at Bragg angles of 34.06° and 65.51° . Chromium $K\alpha$ radiation is used. Determine the crystal structure (if possible) and the lattice parameter. Show that the third and higher reflections cannot occur.

Solution 1: BCC lattice parameter

Given $\sin^2 \theta = 0.120, 0.238, 0.357, 0.475, 0.593, 0.711, 0.830$,

$$\frac{\sin^2 \theta_i}{0.120} = 1.000, 1.983, 2.975, 3.958, 4.942, 5.925, 6.917, \\ \approx 1 : 2 : 3 : 4 : 5 : 6 : 7.$$

The near-integer sequence includes 7. SC skips $N = 7$, while FCC and DC do not begin $1 : 2$. Thus the crystal is BCC, with $N = 2, 4, 6, 8, 10, 12, 14$, indexed as

$$(110), (200), (211), (220), (310), (222), (321).$$

Using Cu $K\alpha$ radiation, $\lambda = 1.5406 \text{ \AA}$,

$$a_i = \frac{\lambda \sqrt{N_i}}{2 \sqrt{\sin^2 \theta_i}} = 3.145, 3.158, 3.158, 3.161, 3.163, 3.165, 3.164 \text{ \AA}.$$

Result

The crystal is **BCC**, with $\bar{a} = 3.159 \text{ \AA} \approx \boxed{3.16 \text{ \AA}}$.

Solution 2: diamond cubic

For $\theta_1 = 34.06^\circ$, $\theta_2 = 65.51^\circ$, and Cr $K\alpha$ radiation ($\lambda = 2.29 \text{ \AA}$),

$$\sin^2 \theta_1 = 0.3137, \quad \sin^2 \theta_2 = 0.8282, \quad \frac{\sin^2 \theta_2}{\sin^2 \theta_1} = 2.640 \approx \frac{8}{3}.$$

The ratio matches the first two diamond-cubic lines, $N = 3$ (111) and $N = 8$ (220). Assuming all lower-angle allowed lines would be observed, the missing FCC $N = 4$ (200) line makes the pattern consistent with **diamond cubic**.

$$a_{111} = \frac{2.29\sqrt{3}}{2 \sin 34.06^\circ} = 3.541 \text{ \AA}, \quad a_{220} = \frac{2.29\sqrt{8}}{2 \sin 65.51^\circ} = 3.559 \text{ \AA}.$$

The next allowed DC line is $N = 11$ (311). Relative to the second line,

$$\sin^2 \theta_3 = \frac{11}{8} \sin^2 65.51^\circ = 1.139 > 1.$$

Result

$a \approx \boxed{3.55 \text{ \AA}}$. Since $\sin^2 \theta_3 > 1$, the (311) line and every higher line are impossible.

Question from lecture slide 22

The characteristic $K\alpha$ line of copper has a wavelength of $\lambda = 1.5406 \text{ \AA}$.

1. Calculate the energy of a Cu $K\alpha$ photon and the equivalent accelerating potential corresponding to this energy.
2. In order to actually produce Cu $K\alpha$ radiation, an incident electron must first eject a K-shell electron of copper. The K-shell binding energy (K-edge) of copper is approximately 8.98 keV. Calculate the minimum accelerating potential required for this process.
3. Explain why the potential calculated in part (2) is higher than that obtained in part (1).

Constants: Planck's constant $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$, speed of light $c = 3.0 \times 10^8 \text{ m/s}$, and charge of an electron $e = 1.602 \times 10^{-19} \text{ C}$.

Solution 3: Cu $K\alpha$ energy

For $\lambda = 1.5406 \text{ \AA} = 1.5406 \times 10^{-10} \text{ m}$,

$$E = \frac{hc}{\lambda} = \frac{(6.626 \times 10^{-34})(3.0 \times 10^8)}{1.5406 \times 10^{-10}} = 1.2903 \times 10^{-15} \text{ J}.$$

$$\frac{E}{e} = 8.054 \times 10^3 \text{ eV} = 8.054 \text{ keV}, \quad V_{\text{equiv}} = \frac{E}{e} \approx \boxed{8.05 \text{ kV}}.$$

The Cu K-edge is 8.98 keV, so the ideal threshold for ejecting a K-shell electron is

$$eV_{\text{min}} = 8.98 \text{ keV} \Rightarrow \boxed{V_{\text{min}} = 8.98 \text{ kV}}.$$

Why the K-edge voltage is higher

Creating the vacancy costs the full K-shell binding energy. A $K\alpha$ photon carries only the difference between the K- and L-shell energies, about $8.98 - 8.054 = 0.926 \text{ keV}$ less. Practical tubes therefore operate above the threshold.

Thank You

Questions?